JOHN MONASH SCIENCE SCHOOL

Mathematical Methods (CAS) Unit 3
Test 2 Part 1 2014

Name:____

Instructions: No calculators or reference notes allowed.

Total /20 marks

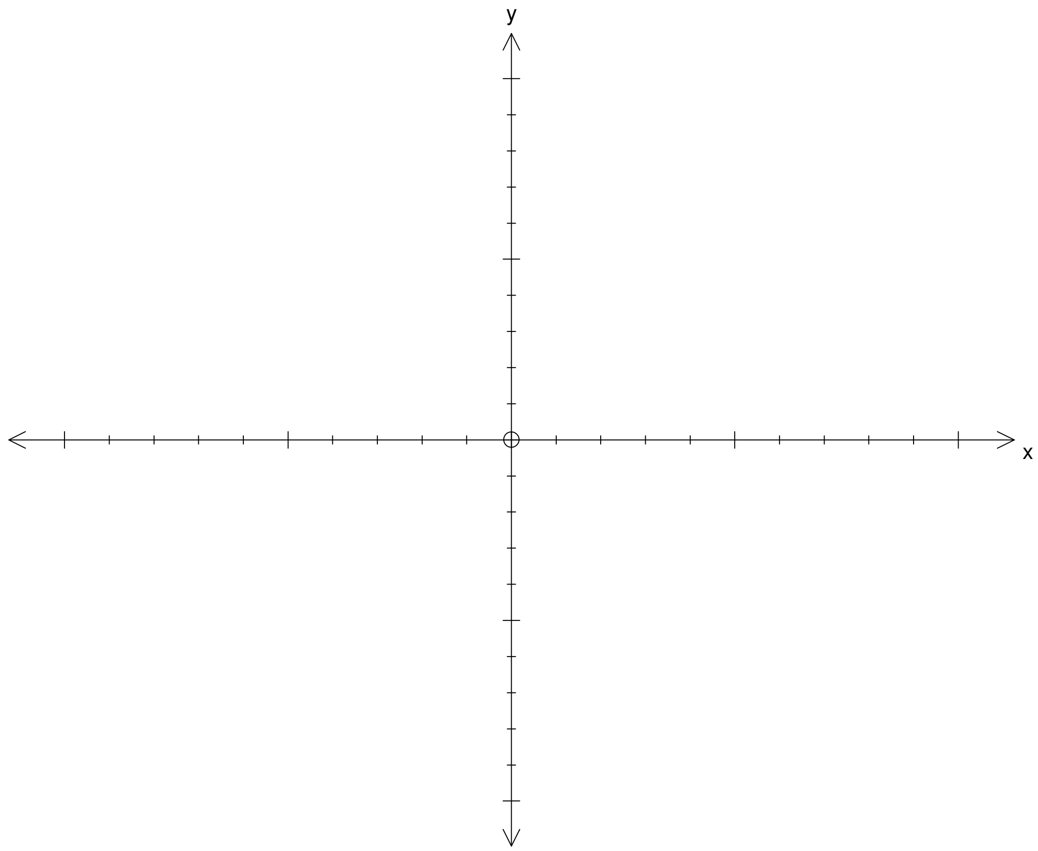
Writing time: 25 minutes

Question 1 (8 marks)

a. State the coordinates of the x-intercepts of the function $y = -x^3 + 2x^2 + 15x$. 2 marks

b. Find the x-coordinates of all stationary points of the function $y = -x^3 + 2x^2 + 15x$. 2 marks

c. On the set of axes below, sketch a graph of the function $f(x) = -x^3 + 2x^2 + 15x$, labelling the coordinates of all axes intercepts and **at least one** stationary point. 2 marks



d. Find the equation of the normal at $x = 0$. 2 marks

Question 2 (9 marks)

Consider the function defined by the equation $y = 1 - \log_2(8 - 5x)$.

a. Find the maximal domain. 1 mark

b. Find in simplest exact form the coordinates of the:

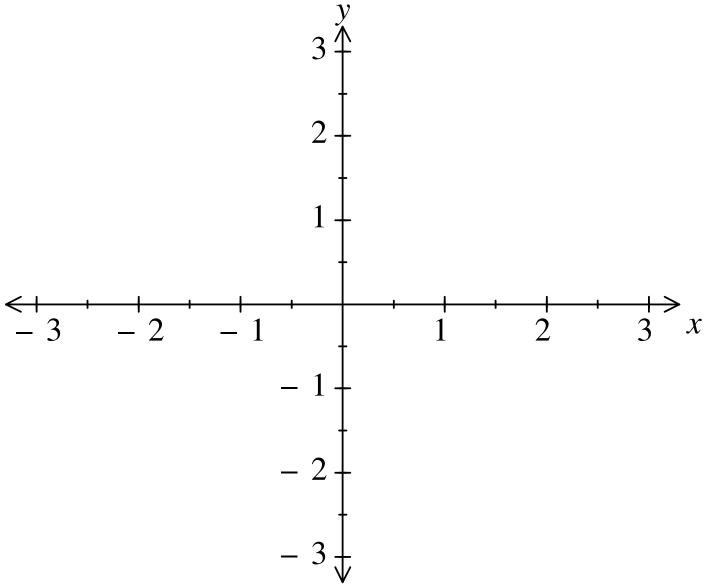
i. y-intecept. 1 mark

ii. x-intercept. 2 marks

c. State the equation of all asymptotes. 1 mark

d. Find the rule for the inverse function. 2 marks

e. On the set of axes below, sketch a graph of the **inverse function**, clearly labelling all important features. 2 marks



Question 3 (3 marks)

Find all solutions to $4^x - 6(2^x) = 16$.

END OF PART 1

JOHN MONASH SCIENCE SCHOOL

Mathematical Methods (CAS) Unit 3
Test 2 Part 2 2014

Name:____

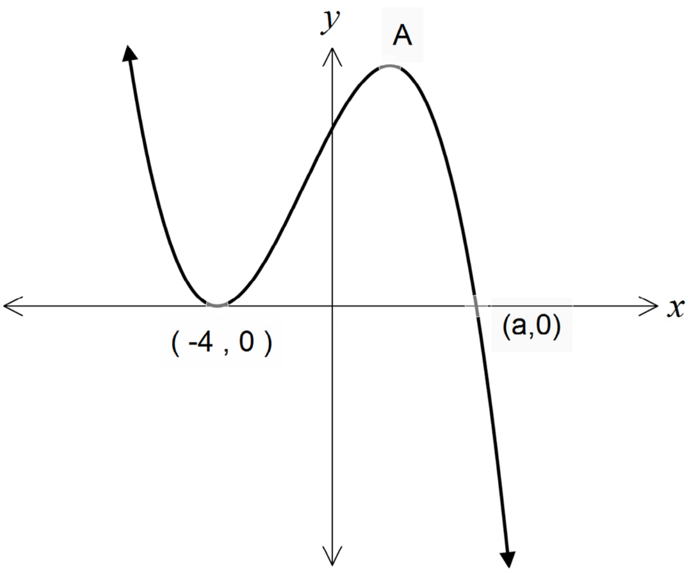
Instructions: CAS calculators are permitted. No reference notes allowed.

Total /7 marks

Writing time: 5 minutes

Question 1

The graph of $y = (x+4)^2(a-x)$ for $a \in (-4,10]$ is shown below.



a. Find $\frac{dy}{dx}$ 1 mark

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b. Find the co-ordinates of the turning point. 2 marks

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c. Find the value of a such that the maximum turning point at A lies on the y-axis. State the coordinates of the maximum turning point. 2 marks

d. For $y = f(x)$, find the exact value of a such that the equation $f(x) = 5$ has exactly two solutions. 2 marks

END OF PART 2